

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING AND
DESIGN

Course Code: FCE 3242

TUTORIAL THREE

DEVELOPMENT OF SURFACES

3(a) Development of a truncated cylinder

A solid cylinder placed with its axis horizontal is truncated by two cutting planes as indicated in the front view shown in Figure Q3(a). One of the cutting planes is inclined at 45° and the other at 30° to the horizontal. Both planes are perpendicular to the Front Picture Plane. The cylinder has a base diameter of 80mm and a length of 100mm. Using a scale of 1:1 and drawing in first angle orthographic projection, draw on separate sheets of paper:

- (i) The Front View, Top View and the Right Side View of the remaining portion of the cylinder.
- (ii) The development of the remaining portion of the cylinder assuming it is now hollow and open on both ends.

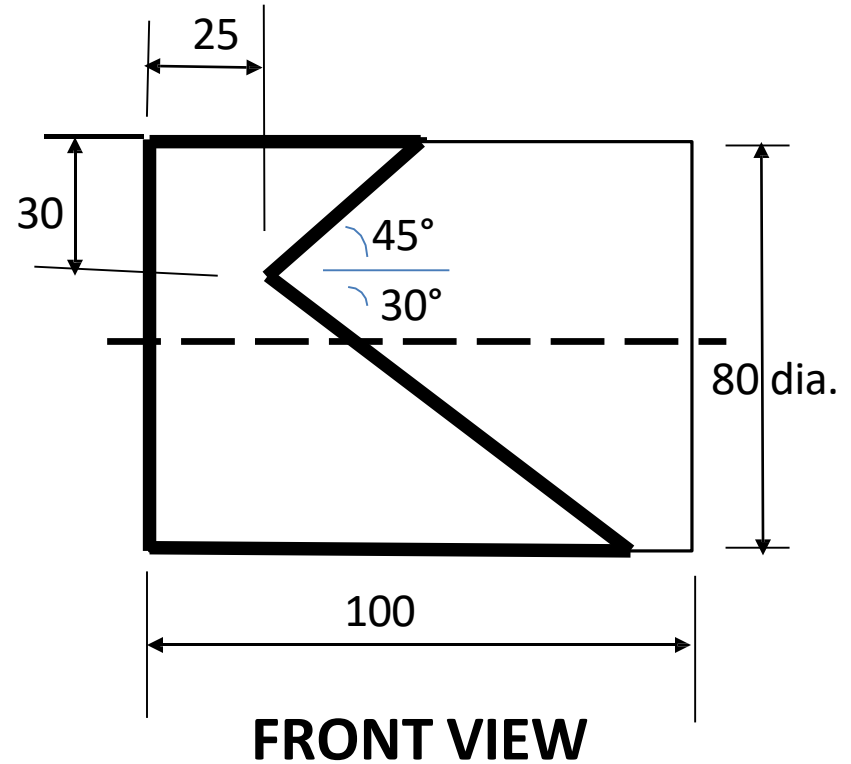


Figure Q4(a)

3(b) Development of a truncated cone

Figure Q3(b) shows the Front View of the portion that remains when a cone has been truncated by two cutting planes as indicated. Each of the cutting planes is perpendicular to the Front Picture Plane. Using a scale of 1:1 and first angle orthographic projection, draw:

(i) The Front View, Top View and the Left Side View of the portion.

(ii) The development of the remaining portion of the cone assuming it is hollow.

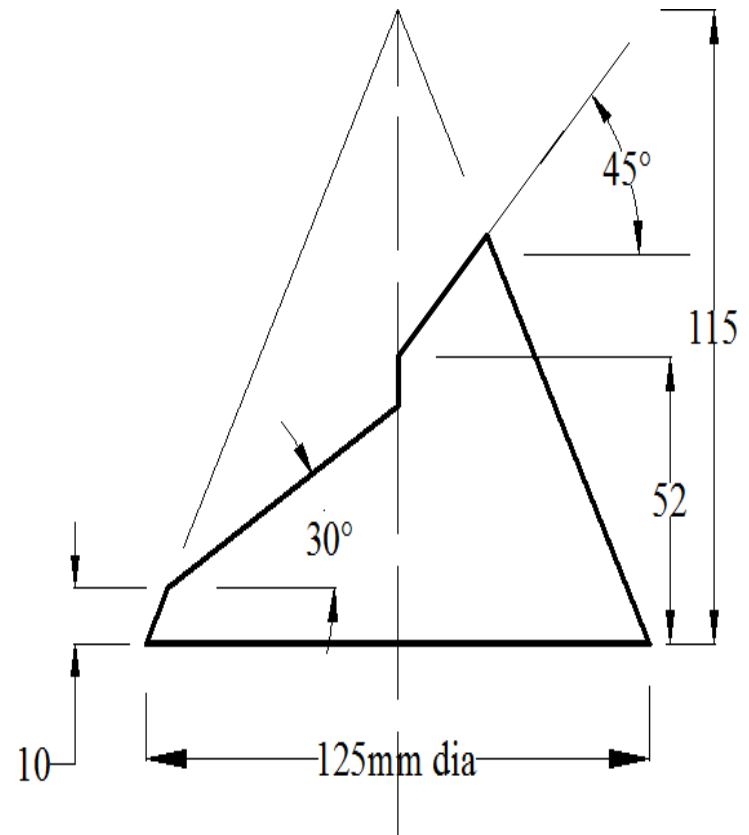


Figure Q3(b)

3(c) Development of a truncated pyramid

A right square pyramid is truncated by two planes that are perpendicular to the Front Picture Plane but inclined to the horizontal as shown in Figure Q3(c) which is the front view of the remaining piece of the pyramid. The height of the pyramid is 120mm and the diagonal of its base is 100mm. Both the top and bottom cut surfaces are covered.

Using first angle orthographic projection and choosing a scale of 1:1,

(a) Draw the Front View, Right Side View and the Top View of the piece.

(b) Draw the development of the remaining piece of the pyramid including the top and bottom covers.

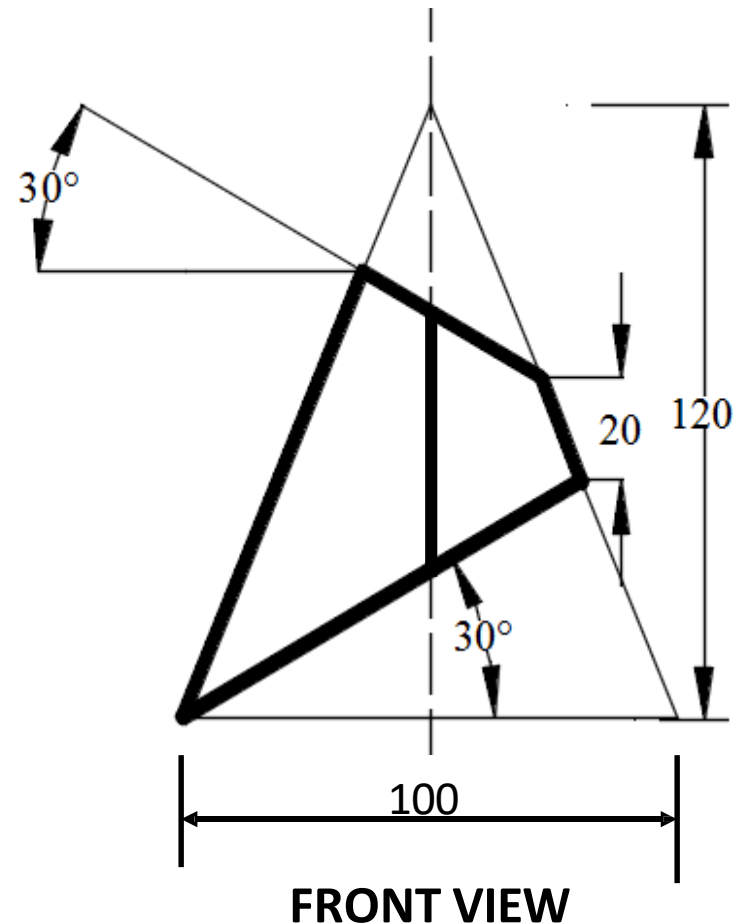


Figure Q3(c)

3(d) Development of a 90°, hexagonal elbow

Figure Q3(d) shows the Front View of a 90° elbow whose bore has a hexagonal cross-section of side 50mm. One side of the elbow is connected to a reducer with a shape of truncated hexagonal pyramid as shown in the figure. The plane which truncates the pyramid is perpendicular to the Front Picture Plane but inclined at 45° to the horizontal and cuts the axis of the pyramid at 70mm from its base.

Using first angle orthographic projection, choose a suitable scale to draw:

- (i) The Front View, Right Side View and Top View of the object.
- (ii) The development of the elbow and the reducer.

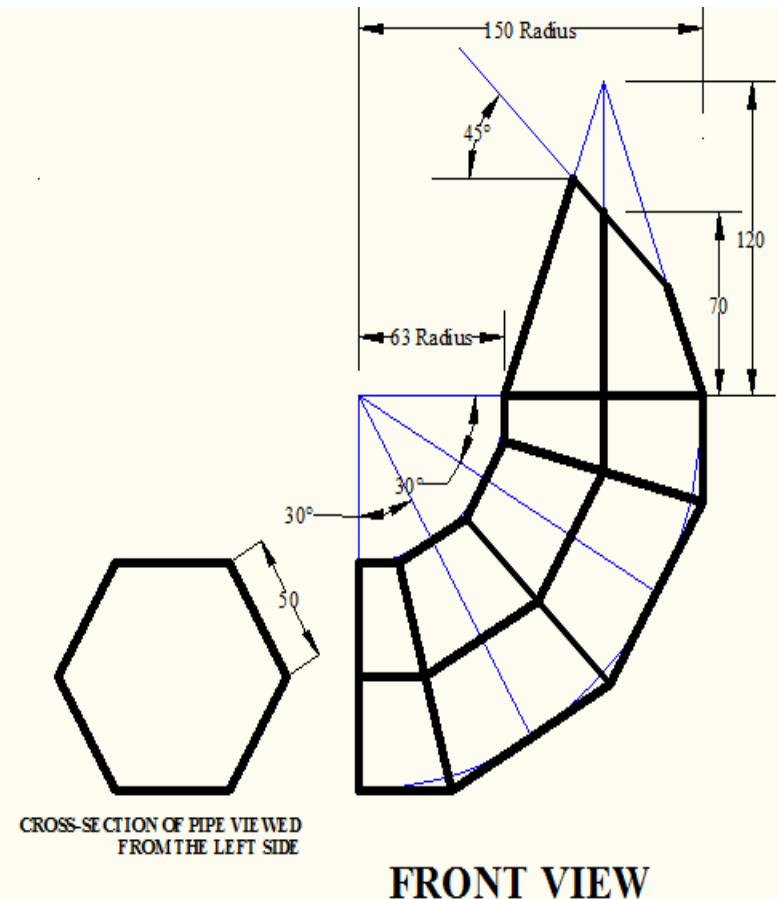


Figure Q3(d)

**TO BE SUBMITTED ON
FRIDAY, 11TH APRIL 2024 AT
2PM**